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11

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

A

A. will likely continue to be used primarily to observe objects outside the solar system.

Choice A is the best answer. The HST will operate until at least 2030, but it's only observing stuff inside our solar system 6% of the time. If we could get a different telescope to observe stuff inside our solar system 100% of the time and take more extensive images of certain things, then the HST could continue to be used mainly for observing stuff outside the solar system.

Choice B is incorrect. This inference is too strong to be supported by the text. Even if the new telescope is deployed, the HST might still be used as it's being used now. Based on the text, the new telescope would just be used for more extensive and long-term imaging of solar system bodies, which doesn't necessarily overlap with the HST. Choice C is incorrect. This inference isn't supported. The text never mentions the possibility of modifying the HST, so there is no basis to make this inference. Rather, the researchers suggest using a different telescope to more closely observe certain objects. Choice D is incorrect. This inference is too strong to be supported. The text doesn't give us enough info to assume that the HST lacks any particular sensors.

Q2 B

B. Polarons allow for the dipoles to synchronize despite higher temperatures.

Choice B is the best answer because it most accurately describes Biliroglu and colleagues' claim about how the polarons function in relation to superfluorescence. The text indicates that "until recently," superfluorescence (intense, synchronized bursts of light emitted by dipoles) has solely been observed at very cold temperatures. However, it also states that, recently, Biliroglu and colleagues report observing the phenomenon at room temperature. They achieved this using "thin films made of perovskite and other similarly crystalline materials," which the researchers claim allows for the formation of polarons. They also suggest that these polarons might absorb the thermal shocks that typically disrupt dipole synchronization at warmer temperatures. Thus, based on the text, Biliroglu and colleagues believe that polarons help dipoles synchronize at temperatures well above those at which superfluorescence had previously been observed.

Choice A is incorrect because the text doesn't address the prospect of a superfluorescent burst moving between crystalline materials or any other mediums. Choice C is incorrect because the text's discussion of polarons is about how they might enable superfluorescence at higher temperatures than those at which it had previously been observed. Rather than suggesting that polarons speed up superfluorescent bursts, the text suggests that no superfluorescence can occur at room temperature in the absence of polarons. Thus, the text indicates that polarons make superfluorescent bursts more likely to occur at higher temperatures than those at which it had previously been observed, not that polarons accelerate the bursts. Choice D is incorrect because the text's discussion of polarons is about how they might enable superfluorescence at higher temperatures than those at which it had previously been observed. In the absence of polarons, the text suggests there would be no superfluorescence at room temperature. Thus, rather than decrease the intensity of superfluorescent bursts, polarons make them more likely to occur under certain circumstances.

Q3 C

C. didn't consistently demonstrate a significant increase in REM sleep after their period of deprivation in the water.

Choice C is the best answer. If REM sleep were homeostatically regulated in fur seals, then all the seals would compensate with REM levels significantly over baseline after going weeks without REM. We'd also expect the seals to maintain those elevated REM levels for some time. Since seals B and C return very quickly to baseline REM levels, this suggests that REM sleep in fur seals may not be regulated homeostatically.

Choice A is incorrect. This doesn't support the conclusion. If REM sleep were homeostatically regulated in fur seals, then we'd suspect the seals to sustain REM levels well above baseline for a prolonged period in order to compensate for weeks of REM deprivation while in the water. Whether or not there's a reduction in REM sleep from day 1 to day 2 doesn't tell us how REM sleep on those days relates to baseline, which is where our focus should be. Choice B is incorrect. The y-axis of this graph doesn't depict baseline levels of REM sleep, but rather shows REM sleep as a percent of baseline. Choice D is incorrect. The graph doesn't depict REM sleep while in the water for the seals in the study. Additionally, we're told fur seals get no REM sleep while in the water, which is significantly different to the values shown in the graph for after they return to land.

Q4

B

B. some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried farming practices with them.

Choice B is the best answer because it presents the conclusion that most logically follows from the text's discussion of Ancestral Puebloans' migration to the Rio Grande Valley. The text states that in the late 1200s C.E., the Ancestral Puebloan civilization abandoned villages in its original homeland, which included the Mesa Verde site. The text goes on to say that recent genetic analysis has demonstrated that the modern turkey population in the Rio Grande Valley descends partly from the ancient turkeys raised at Mesa Verde, and that the genetic markers shared by the two turkey populations first appeared at Mesa Verde only after 1280 C.E. Therefore, it can reasonably be concluded that some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried their agricultural practices—including the farming of turkeys—to their new home.

Choice A is incorrect because the text never compares the condition of the Rio Grande Valley's terrain to that of Mesa Verde's terrain, either in the present or in the past. Choice C is incorrect. Although genetic analysis has demonstrated that the modern turkey population in the Rio Grande valley descended in part from the turkey population raised by the Ancestral Puebloans of Mesa Verde before their migration to the valley in 1280, this finding doesn't eliminate the possibility that Indigenous peoples living in the valley before 1280 might also have farmed turkeys. Choice D is incorrect. The text doesn't consider the possibility that before their migration to the Rio Grande Valley after 1280, the Ancestral Puebloans of Mesa Verde might have adopted turkey farming from an outside Indigenous civilization in another region; instead, the text provides evidence suggesting that the Ancestral Puebloans brought turkey farming to another region—the Rio Grande Valley—after 1280.

Q5 C

C. Although one might expect that foreign investment would increase as natural-resource extraction makes up a larger share of developing countries' economies, the opposite happens because heavy reliance on natural resources can lead to unattractive conditions for investors.

Choice C is the best answer because it accurately states the main idea of the text. According to the text, contrary to what some might expect, foreign investment is typically lower in developing countries whose economies are more dependent on natural-resource extraction. The text explains that high reliance on natural-resource extraction can subject a developing country to economic shocks that can destabilize the local currency and introduce economic uncertainty that tends to keep investors away. In other words, although we may think otherwise, foreign investors are less willing to invest in projects in developing countries whose economies are heavily dependent on natural-resource extraction because those economies tend to exhibit instability that investors want to avoid.

Choice A is incorrect. The text does indicate that foreign investment is typically lower in developing countries whose economies are more dependent on natural-resource extraction; the text further indicates that natural-resource extraction requires substantial initial investments (to acquire things like required technologies) for which there are fewer investors willing to participate at this stage than one might think. But the text does not implicate the cost of these initial investments as a reason why foreign investment is less widely available than some might think. Choice B is incorrect. The text indicates that greater dependence on natural-resource extraction makes a developing country less appealing to foreign investors because of associated economic instability. Rather than arguing that the goal of developing countries is to become less dependent on foreign investment, as the phrasing of choice B suggests, the text focuses only on why foreign investors become less involved with such countries, which suggests that more investment would be preferable. Choice D is incorrect. Although the text indicates that natural-resource extraction requires substantial initial investments (to acquire things like required technologies) and that there are fewer likely investors willing to participate at this stage than one might think, the text does not address what investors are likely to do over time as the industry stabilizes itself.

Q6 A

A. above 344 km/s but below 378 km/s.

Choice A is the best answer because it uses data from the table to give the range of velocities for the LMC from the 1980 value (344 km/s) to the 2006 value (378 km/s), thereby effectively completing the text. The text indicates that before 2006, all the estimated velocities of the LMC were within the range necessary to maintain orbit around the Milky Way galaxy. It then indicates that, according to Besla and colleagues, the 2006 estimate of 378 km/s is too high to maintain that orbit. This strongly implies that if the 1980 value (344 km/s) is below the orbital threshold, and if Besla and colleagues are correct that the 2006 value (378 km/s) is above that threshold, the maximum orbital velocity for the LMC must be somewhere in the range from above 344 km/s to below 378 km/s.

Choice B is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first estimate that exceeded the velocity needed to maintain orbit around the Milky Way. Thus, the 1980 estimate of 344 km/s and the 1994 estimate of 297 km/s must both be below the maximum possible orbital velocity for the LMC. Choice C is incorrect. The text states that Besla and colleagues' analysis found that the velocity from the 2006 study (378 km/s) was too high for the LMC to maintain orbit around the Milky Way. Therefore, if a velocity of 378 km/s is too high, an even higher velocity will also be too high. Choice D is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first to exceed the velocity range required to maintain orbit around the Milky Way. Thus, the 1994 estimate of 297 km/s must be below the maximum possible orbital velocity for the LMC.

Q7 C

C. preserving knowledge about the medicinal value of plants native to the tribal nations' lands.

Choice C is the best answer because it most logically completes the text's discussion of the relationship between Indigenous languages and knowledge of the medicinal uses of plants. The text states that Indigenous cultures possess special knowledge of the medicinal uses of plants, which is reflected in their vocabulary. The text then discusses how tribal nations are working to preserve their languages, whose daily use is declining as globally dominant languages become increasingly dominant in Indigenous communities. Given that the languages of tribal nations in what is now the United States function as repositories of knowledge about plants' medicinal uses, it logically follows that continued use of those languages will assist with passing on knowledge about the medicinal value of plants native to the tribal nations' lands.

Choice A is incorrect because the text states that preserving Indigenous languages will increase the knowledge, not the number, of medicinal plants. Choice B is incorrect because the text is concerned with how vocabulary about the medicinal value of plants can be preserved through the continued daily use of Indigenous languages, not with how such vocabulary can be incorporated into globally dominant, non-Indigenous languages. Moreover, the text explains that the exclusive use of globally dominant languages in Indigenous communities comes at an expense to the continued daily use of those communities' languages. Given this relationship, it is unlikely globally dominant languages would borrow Indigenous vocabulary pertaining to plants' medicinal uses. Choice D is incorrect because the text doesn't discuss physical access to medicinal plants, instead focusing on Indigenous knowledge and language surrounding the medicinal uses of plants.

Q8

B

B. Nature-based approaches can be effective ways to achieve conservation goals.

Choice B is the best answer because it best states the main idea of the text: that nature-based approaches can be effective for achieving conservation goals. The text indicates that in many cases where conservationists are trying to protect ecosystems, their methods depend on natural processes or features. The text then gives an example of this phenomenon, a project with the Quinault Indian Nation that allowed logjams to form naturally in a river, creating spawning habitats for blueback salmon.

Choice A is incorrect. Although the text does suggest that the partnership with the Quinault Indian Nation was beneficial, this is not the central aim of the text; the text primarily argues that nature-based approaches to conservation can be effective. Choice C is incorrect. Although the text indicates that logjams are helpful to blueback salmon, the example of the blueback salmon project is included to illustrate the larger point made earlier in the text: that nature-based approaches to conservation are often effective. Choice D is incorrect. There is no evidence in the text to support a direct comparison of the efficacy of nature-based conservation approaches to other types of approaches. The text merely indicates that nature-based approaches can often be effective.

Q9

B

B. The proportion of municipalities that responded to the inquiry or offered incentives didn't substantially differ across the announcement timing conditions.

Choice B is the best answer. The lighter bars show what happened when the announcement was to come before the election, and the darker bars show what happened when the announcement was to come after the election. For all three of the outcomes, the light and dark bars are virtually the same, demonstrating that the announcement timing didn't actually make a difference.

Choice A is incorrect. This accurately describes some data from the graph, but it doesn't weaken the hypothesis. It doesn't include the "announcement after election" data for comparison. Choice C is incorrect. This accurately describes some data from the graph, but it doesn't weaken the hypothesis. It doesn't include the "announcement after election" data for comparison. Choice D is incorrect. This accurately describes some data from the graph, but it doesn't weaken the hypothesis. It doesn't include the "announcement before election" data for comparison.

Q10 D

D. sunflower sea stars do not always avoid foraging on nutritionally poor sea urchins, making sunflower sea star population recovery a potentially important tool for controlling urchin barrens.

Choice D is the best answer because it presents the conclusion that most logically follows from the text's discussion of the researchers' findings about purple sea urchins and sunflower sea stars. The text explains that urchin barrens are areas that used to be kelp forests but are now covered by purple sea urchins. The text suggests that because there is no more vegetation to be consumed in those areas, the urchins exist in a state of starvation that makes them less nutritional for many predators. The text goes on to explain that in a study with a choice between two purple sea urchins, sunflower sea stars (a predator species that has been substantially declining) consumed a nutritionally rich urchin 42.7% of the time and a nutritionally poor urchin 37.5% of the time. Because the sunflower sea stars didn't always avoid consuming nutritionally poor urchins, even when nutritionally rich ones were available, it follows that helping sunflower sea star populations to grow could help control urchin barrens by increasing the number of sea stars that may consume and thus remove nutritionally poor purple sea urchins from barrens.

Choice A is incorrect because the text indicates only that when presented with purple sea urchins, the sunflower sea stars in the study consumed both nutritionally rich and nutritionally poor ones. It doesn't suggest that sunflower sea stars generally prefer other marine animals that are more nutritious; there's no mention of other marine animals. Choice B is incorrect because the text doesn't suggest that sunflower sea stars are generally reluctant to feed on sea urchins. In fact, the text indicates that the sunflower sea stars in the study did consume sea urchins, feeding on both nutritionally poor and nutritionally rich ones. Choice C is incorrect because the text addresses only the willingness of sunflower sea stars to consume the type of sea urchins found in barrens (nutritionally poor sea urchins), not how likely other species of sea stars are to consume them.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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